

#6 Turn alterations in pill bugs

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1. Purpose

- Observe the alternating turning behavior (turn alternation) in pill bugs

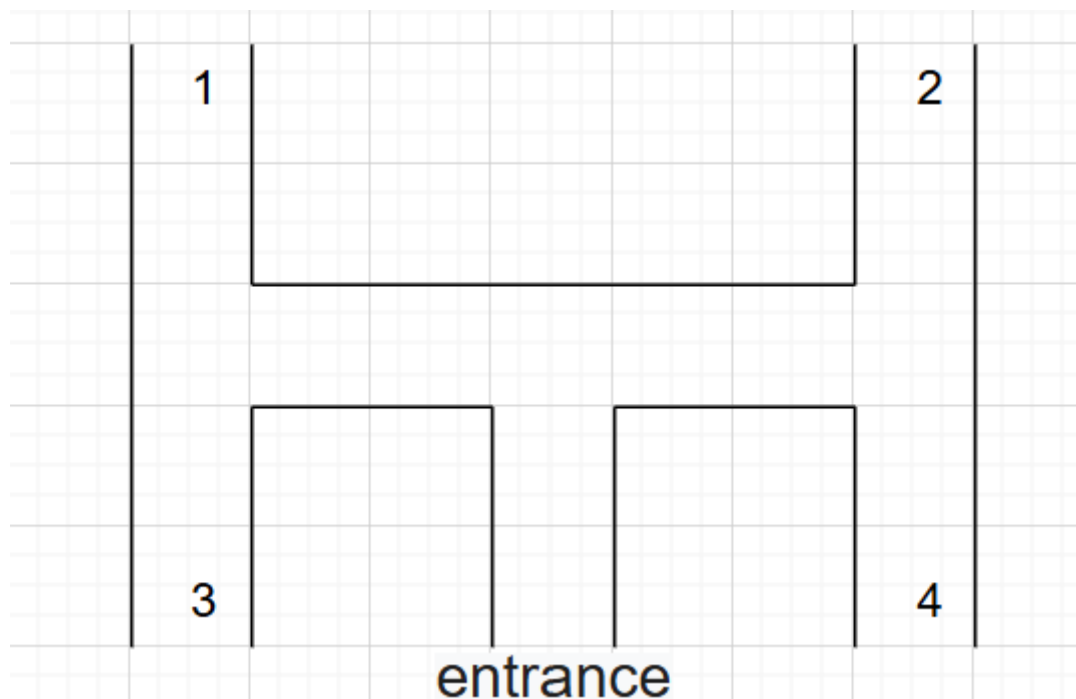
2. Background

- Verify the trait that if a pill bug turns right, it tends to turn left next

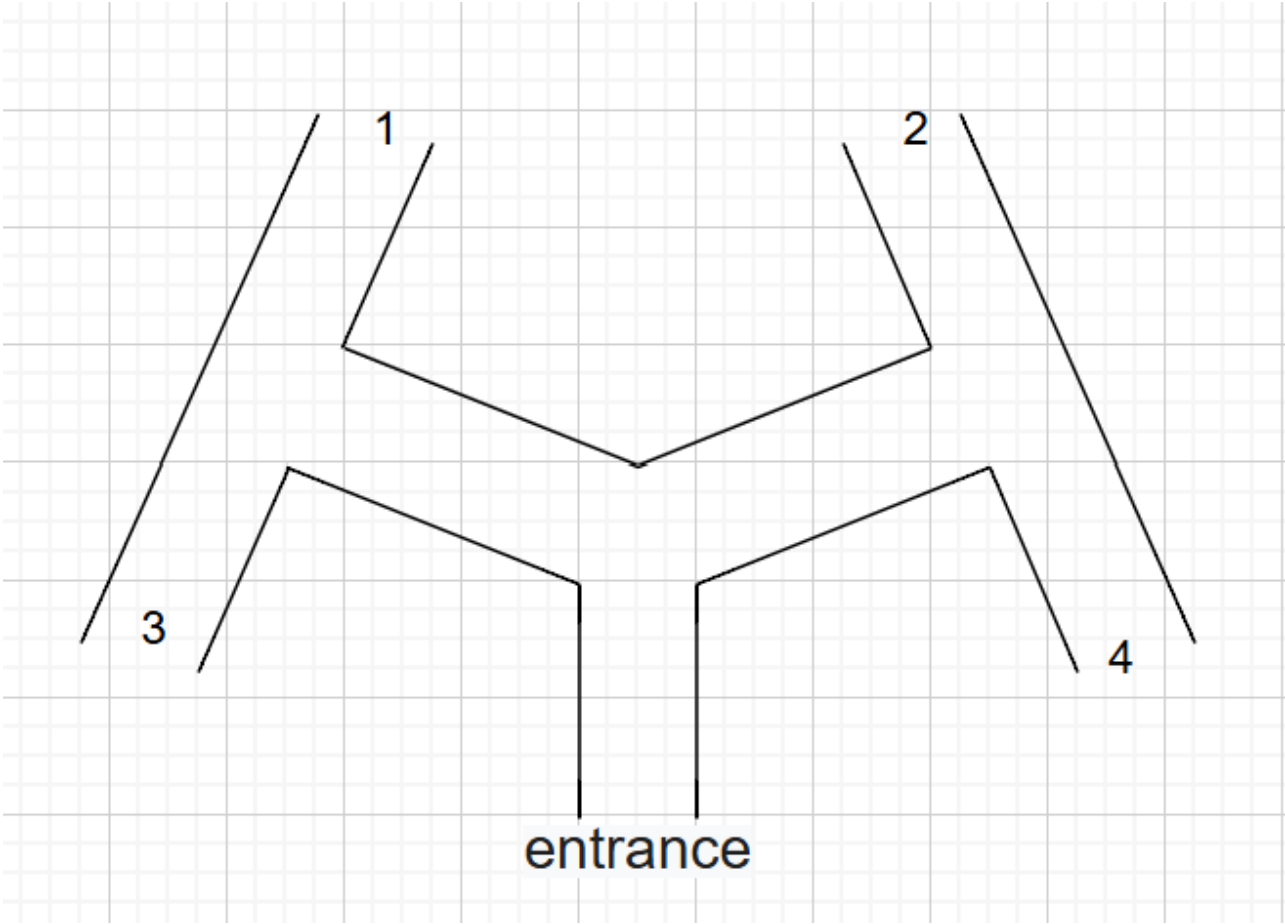
3. Method

- Use an H-shaped (*fig.1)
- Use a Y-shaped(120°) (*fig.2)
- Release a set number of pill bugs and record which direction they turn
- Note number of bugs passing each route

<*fig.1>



<*fig.2>

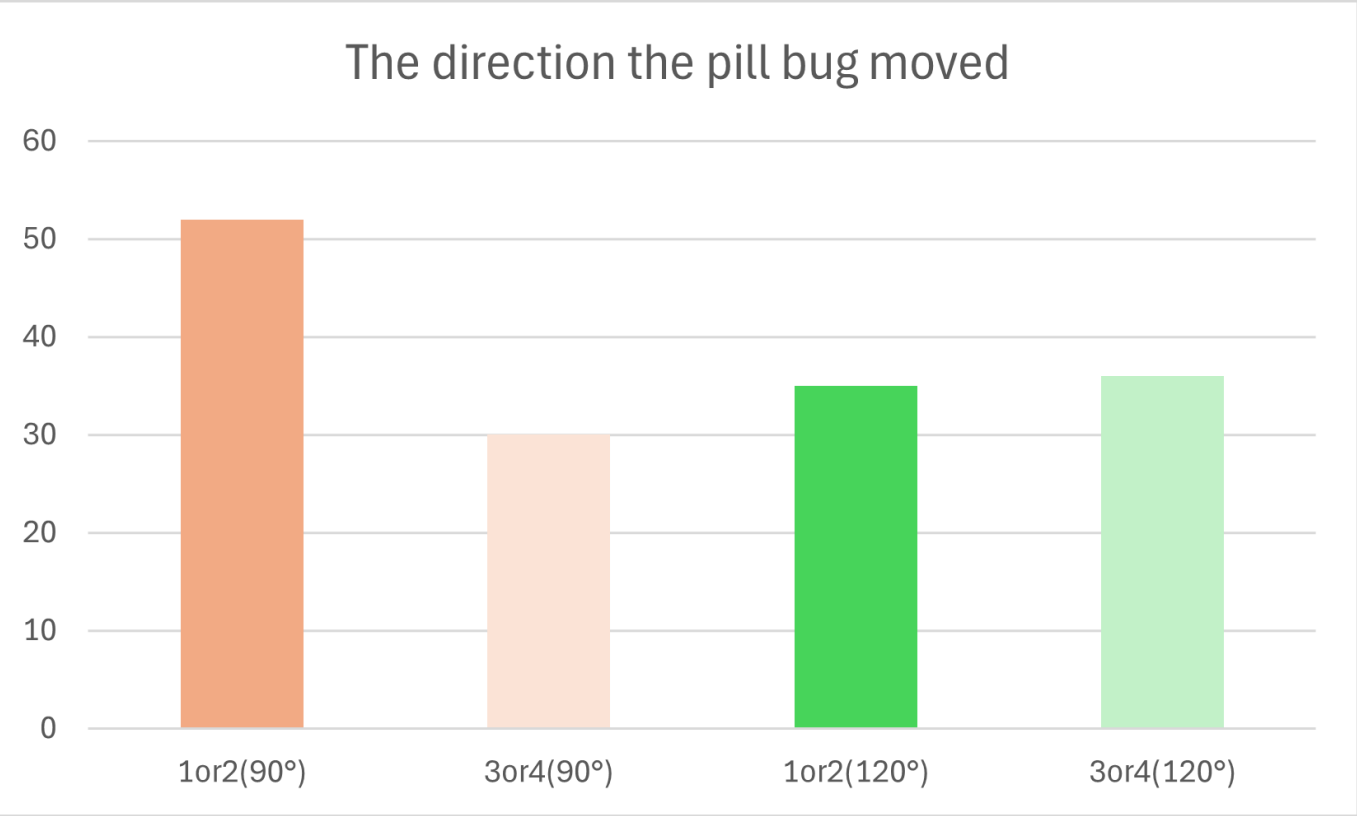


4. Results

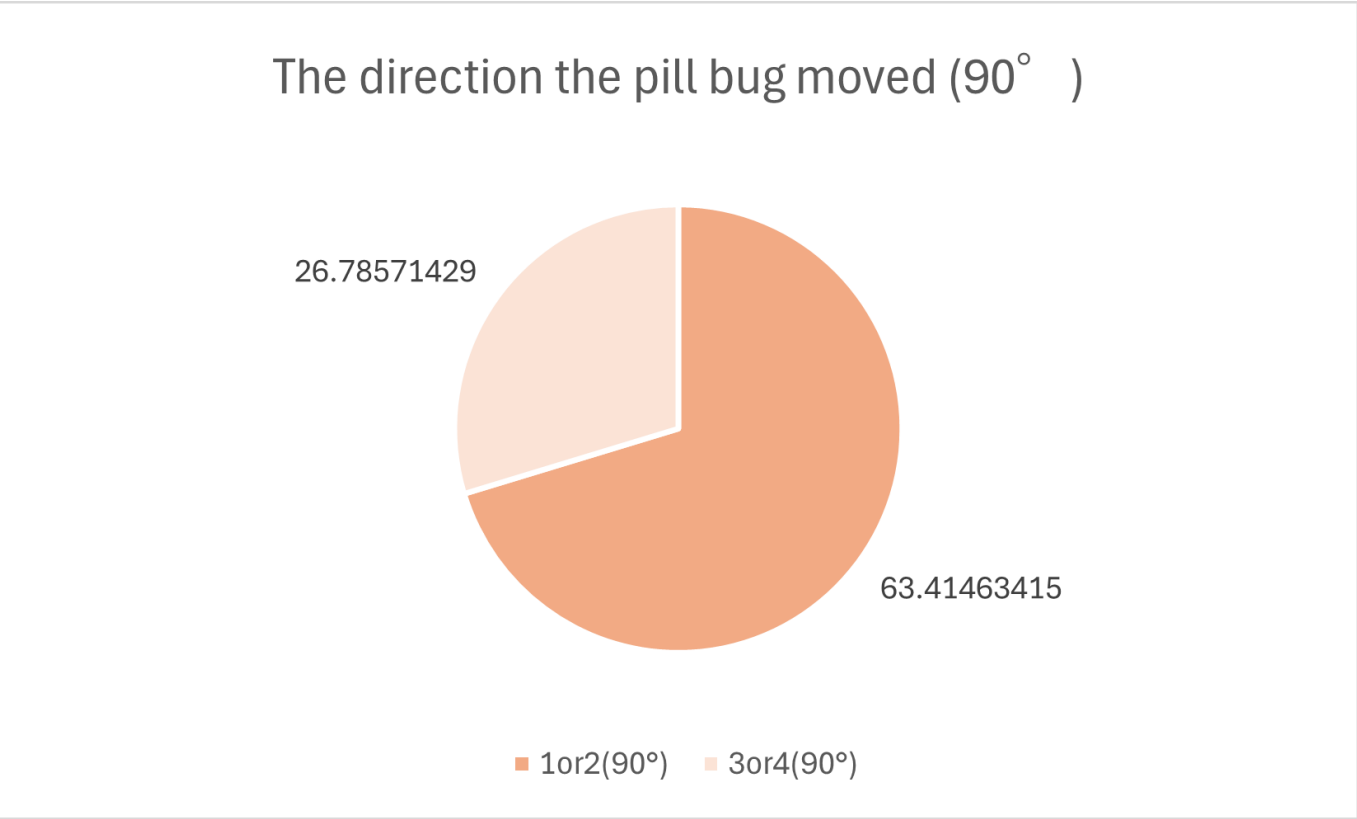
90°	1or2	3or4	Total	120°	1or2	3or4	Total
Class 2	38	18	56	Class 1	14	7	21
Class 4	14	12	26	Class 3	21	29	50
Total	52	30	82	Total	35	36	71
Probability	63.4146341	26.7857143		Probability	49.2957746	50.7042254	

4-1 graph

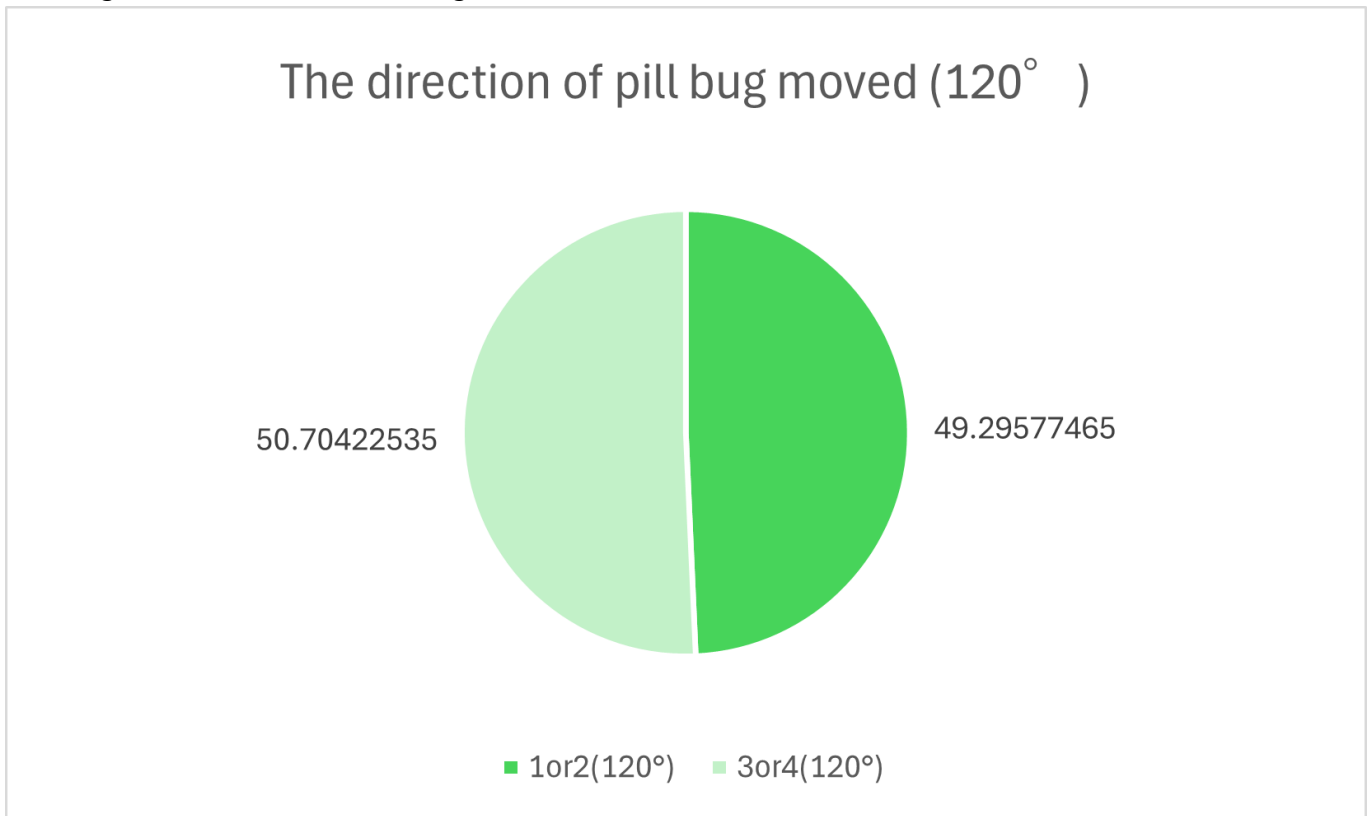
<Comparison of Turn Directions Between 90° and 120° Mazes>



<Pill Bug Movement Pattern (90° Angle)>



<Pill Bug Movement Pattern (90° Angle)>



5. Discussion

In the professor's introduction, we heard that a turn alternation response was observed at a ratio of 9:1. However, in our actual experiment, only about 60% of the turn alternation response was observed.

Possible reasons include the following:

- **Environmental factors related to the passage width**

The passage might have been too wide or too narrow, making the "stimulus from one side," which triggers turning, rather ambiguous.

- **External factors due to pill bugs' aversion to light**

Because pill bugs exhibit negative phototaxis, the unstable light from either side might have caused them to prioritize avoiding light over turn alternation.

Similarly, in this experiment, LEGO blocks were used to construct the path, and the color differences of the blocks might have influenced their behavior.

- **Individual variation and stress factors**

Some pill bugs may have been stressed by handling or the experimental environment, which could have affected their natural turning behavior. Additionally, individual differences in size or age might have contributed to the variation in responses.

- **Timing and observation method**

The speed at which pill bugs moved through the maze and the timing of our observations might not have captured all turning behaviors accurately, especially if some bugs paused or moved very slowly at decision points.

6. Conclusion

As a result, although the ratio was not ideal, a turn alternation response was still observed.

Personally, I found these experimental results very interesting and felt motivated to learn more about animal habits.

It is said that, when humans are undecided between right and left, they choose left 70% of the time. This experiment has given me a good opportunity to research that as well 😊